

Awesome Title

by

Your Name Here

A thesis
submitted to the University of Ottawa
in partial fulfillment of the
thesis requirement for the degree of
Doctor of Philosophy
in
Physics

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Abstract

This is a placeholder abstract. Replace this text with a concise summary of your thesis, including the motivation for the work, the methods used, the main results, and the most important conclusions.

In this thesis, we investigate an extremely important scientific problem using a collection of very sophisticated methods. Several experiments are performed, many equations are solved, and a non-negligible amount of coffee is consumed. The results show that something interesting happens under certain conditions, in reasonable agreement with expectations. These findings provide new insight into the problem and suggest several exciting directions for future work.

Seriously, replace this abstract before submitting your thesis.

Acknowledgments

This is a placeholder acknowledgements section. Replace this text with your own acknowledgements before submitting your thesis.

You may wish to thank your supervisor, committee members, collaborators, laboratory colleagues, funding agencies, friends, family, and anyone else who supported you during your degree.

For example, you could write something thoughtful about your supervisor's guidance, your colleagues' helpful discussions, and your family's support. You could also acknowledge the indispensable contributions of coffee, late-night snacks, and the one piece of laboratory equipment that only works when nobody is watching.

Seriously, though—replace this text. If you are reading this in your submitted thesis, it is already too late.

Research is what I'm doing when I don't know what I'm doing.
Placeholder Researcher

List of Publications

1. A. N. Author, **S. T. U. Dent**, and P. H. D. Advisor, “A very important investigation of an extremely interesting physical phenomenon.” *Journal of Placeholder Physics* **42**, 100–110 (2023).
2. **S. T. U. Dent**, A. N. Author, and P. H. D. Advisor, “Experimental observation of something that was theoretically expected to happen.” *International Journal of Dummy Research* **17**, 123456 (2024).
3. **S. T. U. Dent**, C. O. Author, A. N. Author, and P. H. D. Advisor, “Further studies on the propagation of light, knowledge, and uncertainty.” *Proceedings of Imaginary Optics* **99**, 200–208 (2025).

Statement of originality and collaborative contributions

To the best of the author's knowledge, the work described in this thesis constitutes original research in the field of physics. The collaborative contributions of the authors to the work presented in each chapter are described below.

Supervisor 1 and Supervisor 2 conceived the idea for the work presented in Chapter [2](#), concerning the study of good photons. Author 1 performed the numerical simulations and analyzed the resulting data. Author 1 and Supervisor 2 prepared the manuscript with contributions from all authors. All authors discussed the results and contributed to the final version of the manuscript.

Supervisor 1 and Author 2 conceived the idea for the work presented in Chapter [3](#), concerning stars on waveguides and weak measurements. Author 1 and Author 2 designed the optical fields. Author 1 performed the numerical simulations, while Author 1 and Author 2 analyzed the resulting data. Author 1, Author 2, and Supervisor 1 prepared the manuscript with contributions from all authors. All authors discussed the results and contributed to the final version of the manuscript.

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Chapter 1

Introduction

Light is an electromagnetic phenomenon. It can behave as a wave and can also be described in terms of particles known as photons. These properties form the basis of classical and quantum optics.

This is placeholder text for the introduction of the thesis. Replace this section with an introduction to the relevant field, the motivation for the research, and an overview of the work presented in the following chapters.

Chapter 2

Light and Focusing

This chapter is based on the following manuscript:

1. **Author 1**, Author 2, Supervisor 1, “Theoretical analysis of light and focus”, *Optics Expresso* **28**, 20026-20028 (2026) DOI: <https://doi.org/>

In the past decades, the use of structured light has revolutionized the field of light and structures. In this chapter, we present a detailed theoretical analysis The formalism used to describe focusing is discussed [2.1](#). The numerical implementation used in this work can be found in Appendix [A](#). We believe the results can be used for something.

2.1 Focused Fields

There was a light beam and it got focused! The experimental implementation is shown in [2.1](#)

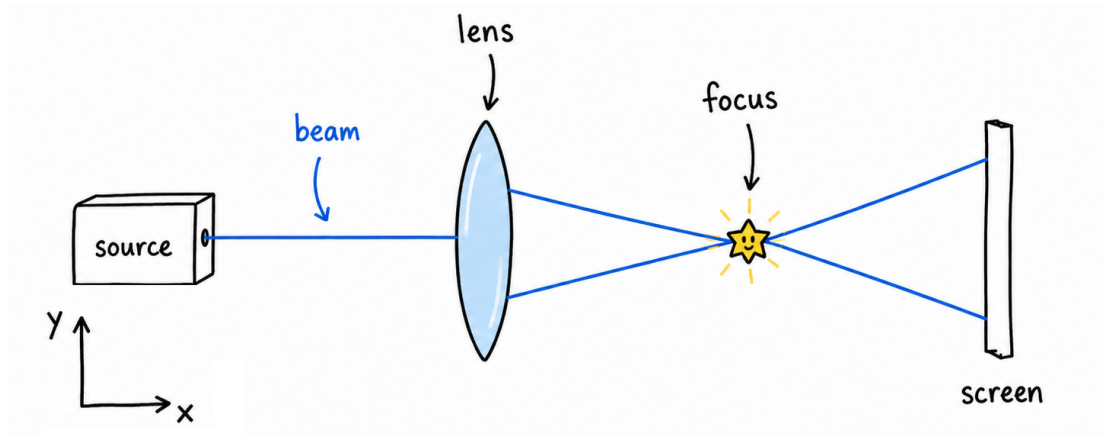


Figure 2.1: **Focusing of optical beams:** a. Schematic diagram of the focusing of light

Some communications

Dummy page

Experimental demonstration of focusing light

Author 1, Author 2, Supervisor 1

Focusing light is easy until the beam decides otherwise. Here, we investigate the familiar process of moving a lens back and forth until a blurry spot becomes slightly less blurry. Using a laser, a lens, and a generous amount of patience, we identify a position that looks reasonably close to the focus. Further adjustments confirm that the previous position was probably better. These findings highlight the importance of careful alignment, stable mounts, and knowing when to stop touching the experiment.

Light generally travels where it is supposed to, except when an experiment requires it to do so. A lens provides a convenient way to bring a beam to a focus, provided that the lens is facing the right direction and has not been borrowed from a neighbouring setup. In practice, the alignment procedure consists of making small adjustments while confidently announcing that the spot is getting smaller.

To locate the focal plane, we placed a screen at several positions along the optical axis. The screen was then moved slightly forward, slightly backward, and finally returned to approximately where it started. The smallest observed spot was recorded as the best result, although a colleague suggested that it could probably be improved.

Repeated measurements showed reasonable agreement whenever nobody touched the table. Future work will investigate whether the remaining imperfections arise from aberrations, mechanical drift, or the mysterious adjustment made just before lunch.

Chapter 3

Waveguiding and Weak Measuring

This chapter is based on the following papers:

1. **Author 1**, Author 2, Author 3, and Supervisor,
“Weak Measuring and Waveguiding”, *Physical Review Re-reseach* **1**, 0326 (2026).
DOI: <https://doi.org/>

Motivation to the chapter

3.1 Wave guiding

Light gets guided in a waveguide. Look at our diagram [3.1](#).

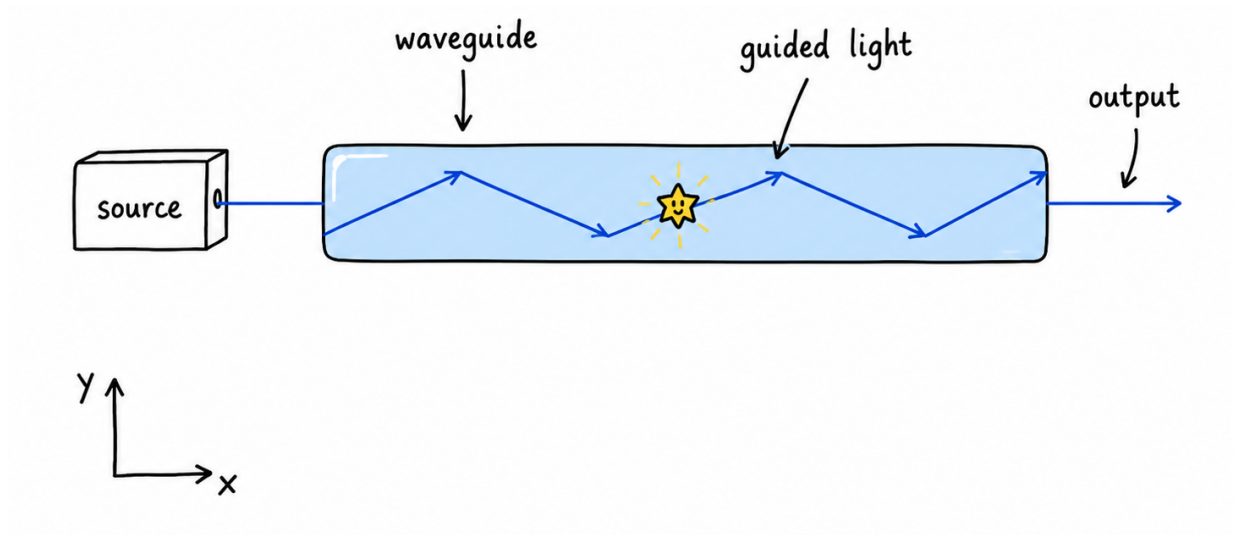


Figure 3.1: **Wave and their guidance.** The photon gets trapped and guided towards somewhere.

3.2 Weak Measuring

If you barely perturb the system, it stays almost the same! Figure [3.2](#) shows how not to do a weak measurement

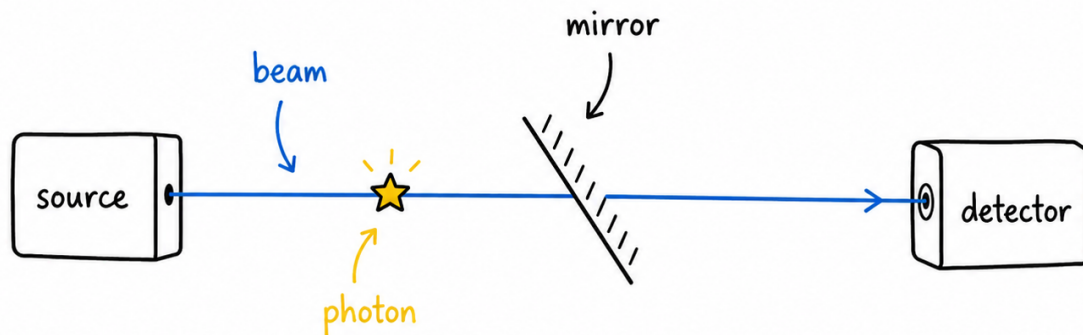


Figure 2. Simple placeholder optical sketch.

Figure 3.2: **Example of a bad weak measurement.** a. The photon sometimes reflects or goes through. Although, I forgot to measure it.

Some communications

Dummy page

Experimental demonstration of focusing light

Author 1, Author 2, Supervisor 1

Focusing light is easy until the beam decides otherwise. Here, we investigate the familiar process of moving a lens back and forth until a blurry spot becomes slightly less blurry. Using a laser, a lens, and a generous amount of patience, we identify a position that looks reasonably close to the focus. Further adjustments confirm that the previous position was probably better. These findings highlight the importance of careful alignment, stable mounts, and knowing when to stop touching the experiment.

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Repeated measurements showed reasonable agreement whenever nobody touched the table. Future work will investigate whether the remaining imperfections arise from aberrations, mechanical drift, or the mysterious adjustment made just before lunch.

Chapter 4

Conclusions

In this thesis, several aspects of light and optical systems were investigated using theoretical, numerical, and experimental methods. The results presented throughout the different chapters illustrate how fundamental concepts in optics can be applied to study a variety of physical systems.

The work discussed here provides a general example of how a thesis conclusion may summarize the main results, highlight their significance, and identify possible directions for future research.

This is placeholder text. Replace this section with a concise summary of the main contributions of your thesis, the conclusions supported by your results, and any relevant future perspectives.

APPENDICES

Appendix A

Numerical calculation of something

In this section, we provide the ... notebooks used for the calculation of...

A.1 Code 1: Mathematica version

A.1.1 Main code

Definition

```
(* Plot something easy *)
intensity[x_] := Exp[-x^2]

Plot[intensity[x], {x, -3, 3}]
```

A.2 Python

A.2.1 Main code

```
import numpy as np
import matplotlib.pyplot as plt

def intensity(x):
    return np.exp(-x**2)
```

```
x = np.linspace(-3, 3, 300)
plt.plot(x, intensity(x))
plt.show()
```

Appendix B

Supplementary Materials: SM1

Supplementary Material 1

The beam was aligned by adjusting the lens position until the spot appeared sufficiently small. To ensure reproducibility, the procedure was repeated several times, usually after someone accidentally moved a component.

Additional measurements were performed by placing a screen before and after the estimated focal plane. As expected, the spot became larger in both directions, confirming that the smallest spot was somewhere in between. The final lens position was marked with tape to discourage further improvements.

Appendix C

Supplementary Materials: SM2

Supplementary Material 2

The stability of the setup was evaluated by observing the focused spot over time. Under normal conditions, the beam remained approximately where it had been left. Small fluctuations appeared when someone leaned on the table or adjusted a component described as unrelated to the experiment.

To reduce these disturbances, measurements were performed while everyone stood still and avoided making helpful suggestions. The alignment remained satisfactory until the next morning, when the procedure described in Supplementary Material 1 was repeated.

References